Create a Factorial API Endpoint and Test It Using Postman

1. Create an ASP.NET Core Web API Project : Create a new project → ASP.NET Core Web API → Next → Project Name: FactorialAPI → Select latest available .NET version → Create

2. Create the API Controller: Solution Explorer → Controllers → Right-click → Add → Controller → API Controller - Empty → Name: FactorialController.cs

3. Write the Factorial API Code

```
using Microsoft.AspNetCore.Mvc;
namespace FactorialAPI.Controllers
{
    [ApiController]
    [Route("api/[controller]")]
    public class FactorialController : ControllerBase
    {
        [HttpGet("{number}")]
        public IActionResult CalculateFactorial(int number)
        {
            if (number < 0)
            {
                return BadRequest("Factorial is not defined for negative numbers.");
            }
            long factorial = 1;
            for (int i = 1; i <= number; i++)
            {
                factorial *= i;
            }
            return Ok(new
            {
                Number = number,
                Factorial = factorial
            });
        }
    }
}
```

4. Understand the API Endpoint

```
[Route("api/[controller]")]
[HttpGet("{number}")]
```

If the application is running at https://localhost:7001 and we want to calculate the factorial of 5, the API URL is:

```
https://localhost:7001/api/factorial/5
```

The request uses the GET HTTP method.

5. Run

6. Test the API Using Postman

```
https://localhost:7001/api/factorial/5
```

7. Check the Response

For the number 5, Postman should display:

```
{
  "number": 5,
  "factorial": 120
}
```